

Dr. Bushra Naz

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Professional Summary

Distinguished and forward-thinking engineering professor recognized for pioneering contributions to Artificial Intelligence, Machine Learning, Deep Learning, and Image Processing. A proven academic leader and visionary researcher with extensive expertise in Artificial Intelligence, image classification, object detection, and advanced pattern recognition. Renowned for driving high-impact interdisciplinary research collaborations that bridge academia, industry, and global innovation ecosystems. Adept at mentoring and developing future leaders at the PhD, Master's, and undergraduate levels while securing competitive research funding and delivering transformative solutions. Committed to advancing responsible, scalable, and cutting-edge AI applications in digital image processing, satellite image analysis, computer vision, and multimodal intelligent autonomous systems to address complex real-world challenges.

Professional Experience

Associate Professor (BPS-20), Department Of Computer Systems Engineering, MUET, Pakistan (Jan 2022 – Present)

Assistant Professor (BPS-19), Department Of Computer Systems Engineering, MUET, Pakistan (Nov 2018 – Jan 2022)

Lecturer (BPS-18), Department Of Computer Systems Engineering, MUET, Pakistan (Feb 2009 – Oct 2018)

Lab Lecturer (BPS-18), Department Of Computer Systems Engineering, MUET, Pakistan (Feb 2008 – Jan 2009)

Internee, Department Of Computer Systems Engineering, MUET, Pakistan (Aug 2007 – Feb 2008)

Education

PhD in Computer Science and Engineering (2014 – 2018)

Nanjing University Of Science & Technology, Nanjing, China

Thesis: Spectra-spatial Elastic-Net Representation Based Methods for Hyperspectral Image Classification

Master's Degree in Communication System & Networks (2007 – 2009)
Bachelor's Degree in Computer Systems Engineering (2003 – 2007)

Research Grants (Principal Investigator / Lead)

- 2023 – ICT Endowment Fund: Multimodal Speech Emotion Recognition Using Deep Neural Network
- 2023 – NGIRI: Indoor Guidance Smart Glasses for the Blind (Final Year Project)
- 2023 – NGIRI: Augmented Reality Based Outdoor Navigation System (Final Year Project)
- 2022 – NGIRI: IoT-based Water Quality Monitoring with Bacterial Prediction (Final Year Project)
- 2021 – NGIRI: Vehicle Accident Detection and Information System using Android & GPS
- 2019 – HEC SRGP: Crop Sensing & Monitoring via Multitemporal Distinguishability and Classification

Research Supervision

- Principal Supervisor: Kalsoom Panhwar, PhD (Completed): Title: **Optimizing Transformer Based Deep Learning Model for classification of Desertification through Satellite Images**
- Principal Supervisor: Multiple BE and ME research projects
- Co-Supervisor: Aisha Ahmed, PhD (Ongoing): Mamba-AVGard: An Optimized Audio-Visual Deepfake Detection Framework Using Mamba Vision Transformers

Awards & Honours

- China Scholarship Council PhD Award (2014–2018)
- ELITE Best Foreigner Student Scholarship (2017)
- HEC Scholarship for ME Degree (2008–2009)
- UNESCO/PRC Fellowship Senior Research Scholar (2011–2012)
- IEEE JSTARS Best Paper Honor Certificate (2017)
- Merit Scholarship for BE (2003–2006)

Research Interests

- Image Processing, Classification, Object Recognition
- Remote Sensing Image Analysis, Sparse Representation
- Temporal and Multimodal Data Analysis
- AI and Machine Learning Optimization Methods
- Hyperspectral Imaging and Data Mining
- Signal Processing and Statistical Modeling

Professional Affiliations & Activities

- IEEE Member, PEC Member, Certified Program Evaluator since 2025
- Session Chair: INCCST'22 (2022), ICETECC (2023), INCCST'19 (2019)
- Reviewer: IEEE JSTARS, TGRS, Pakistan J. of Computer & Information Systems
- Reviewer: Journal of Communication (China), IEEE ISEC 2021, ICOSST 2022
- Committee Member: ICOSST 2022

Book Chapter

EEG Based Major Depressive Disorder (MDD) Detection Using Machine Learning, January 2022. DOI: 10.1007/978-3-031-04112-9_13, In book: Pattern Recognition and Artificial Intelligence, Nayab Bashir, Sanam Narejo, Bushra Naz, Asif Ali

Peer-Reviewed Journal Publications

- [2026] Maria Saleem, Bushra Naz Soomro, Almas Ali, Zartasha Baloch, "Multi-Modal Bio signal Transformer Based Deep Learning Approach for Mental Stress Detection", Mehran University Research Journal of Engineering and Technology, IF:0.7, Vol. 45 No. 1 (2026)
- [2025] Panhwar, K.; Soomro, B.N.; Bhatti, S.; Jaskani, F.H. Novel Encoder– Decoder Architecture with Attention Mechanisms for Satellite-Based Environmental Forecasting in Smart City Applications. *Future Internet* 2025, 17, 380. <https://doi.org/10.3390/fi17090380>.
- [2024] Ali, S., Naz, D. B., Narejo, D. S., & Ahmed, Z. (2024). "Alex Net-Based Speech Emotion Recognition Using 3D Mel- Spectrograms". *International Journal of Innovations in Science & Technology*, 6(2), 426–433. Retrieved from <https://journal.50sea.com/index.php/IJIST/article/view/736>
- [2024] Kulsoom, Naz. B, Bhatti. S, "A Systematic Review of Desertification Identification with Multispectral LANDSAT Image and Deep Learning Models", *IJIST*, Vol. 6 Issue. 1 pp 143-169, Feb 2024.
- [2024] Memon. F, Naz. B, Narejo. S, Panhwar. K, "An artificial intelligence Vision transformer model for classification of bacterial colony", *IJIST*, Vol. 6 Issue.1 pp237-248, March 2024.
- [2023] Bushra Naz, Shahzad Hyder, Azlan Ahmed, Ali Hasnain, "A Machine Learning Framework for E. coli Bacteria Detection and Classification", *UMT Artificial Intelligence Review (UMT-AIR)* Y category, Volume 3 Issue 1, pp-18-30, June 2023 DOI: <https://doi.org/10.32350/umtair.31.02>.
- [2023] Bashir, N., Narejo, S., Naz, B. et al. A Machine Learning Framework for Major Depressive Disorder (MDD) Detection Using Non-invasive EEG Signals. *Wireless Pers Commun* (2023). <https://doi.org/10.1007/s11277-023-10445-w>
- [2023] Duolin Wang, Yadang Chen, Bushra Naz, Le sun, Baozhu Li, "Spatial-Aware Transformer (SAT): Enhancing Global Modeling in Transformer Segmentation for Remote Sensing Images", *Remote Sensm.(W category, IF. 5.6)* 2023, 15(14), 3607. <https://doi.org/10.3390/rs15143607>.
- [2023] Shabana Hajano, Dr. Bushra Naz, and Dr. Shanwaz Talpur, Sofia Hajano, Classification of Amputee EMG Signals using Machine Learning Techniques, *International Journal of Innovations in Science & Technology*, Y category, 2023.
- [2023] Mahnoor Masroor, Bushra Naz, Shahnawaz Talpur, Zohaib Ahmed, Sparse Deep Learning & Hyper Spectral Image Classification, *engineering Science and technology international research journal*, Vol.7, No, 3, Sep: 2023.

- [2023] Syed Abdul Hanan Ahmed Shah, Shahnawaz Talpur, Bushra Naz, "A Discriminative Image Registration Model with Hybrid Feature Detector", International Journal of Scientific Research in Computer Science, Engineering, and Information Technology, vol 9 (3), 2023, pp:272-278, doi: <https://doi.org/10.32628/CSEIT2390369>.
- [2022] Nuzhat Faraz, Bushra Naz, Sheeraz Memon, "Data Mining Approach for Detection and Classification of Brain Tumor", Mehran University Research Journal of Engineering and Technology (Y category). Vol.41, No. 1, 52 - 63, 01 January 2022.
- [2022] Nayab Bashir, Sanam Narejo, Bushra Naz, Mohammad Moazzam Jawed, Shahnawaz Talpur, Khurshid Aliev "Non-invasive EEG based Feature Extraction framework for Major Depressive Disorder analysis", International Journal of Innovations in Science & Technology (Y category), ISSN2618-1630, Vol.4 No.1 (2022), pp. 110-122
- [2022] Zohaib Ahmad, Bushra Naz, Dr. Bhawani Shankar, Sara Ali, Zakir Shaikh, "Deep Feature Learning And Classification Of Remote Sensing Images", UMT Artificial Intelligence Review (UMT-AIR) (Category: Y), Vol 2 No 1 (2022), <https://doi.org/10.32350/umtair.21.004>.
- [2022] Bakhtawer, Bushra Naz , Sanam Narejo, Naseer U Din, Waqar Ahmed, "Soil Classification & Prediction of Crop Status with Supervised Learning Algorithm: Random Forest", International Journal of Innovations in Science & Technology (Y category), Vol 04 Issue 04: pp 1011-1022, 2022.
- [2022] Bibi. A, Naz. B, Talpur. S, Soomro. S. H, Bablani. Y "A Qualified review of ML and DL algorithms for Bearing Fault Diagnosis", International Journal of Innovations in Science & Technology (Y category), Vol 04 Issue 04: pp 998-1010, 2022.
- [2021] Asma Fahim, Qingmei Tan, Bushra Naz, Quratul Ain and Sibghat Ullah Bazai, "Sustainable Higher Education Reform Quality Assessment Using SWOT Analysis with Integration of AHP and Entropy Models: A Case Study of Morocco", Sustainability (W category: IF: 2.576), Volume 13, Issue 7, April 2021, pp-4312.
- [2021] Naseer U Din, Bushra Naz, Samer Zai, Bakhtawer, Waqar Ahmed, "Onion Crop Monitoring with Multispectral Imagery using Deep Neural Network", (IJACSA) International Journal of Advanced Computer Science and Applications (Scopus(Elsevier), Web of Science (Clarivate/Thomson Reuters), Inspec (IET)), Vol. 12, No. 5, 2021, pp-303-309.
- [2021] Asmaa Fahim, Qingmei Tan, Mouna Mazzi, Md Sahabuddin, Bushra Naz, and Sibghat Ullah Bazai, "Hybrid LSTM Self- Attention Mechanism Model for Forecasting the Reform of Scientific Research in Morocco", Hindawi, Computational Intelligence and Neuroscience (W category: IF: 2.284), Volume 2021, Article ID 6689204, 14 pages. <https://doi.org/10.1155/2021/6689204>.
- [2021] Urooj Kaim khani, Bushra Naz, Sanam Narejo, "Rainfall Prediction Using Time Series Nonlinear Autoregressive Neural Network", SSRG International Journal of Computer Science and Engineering, EISSN 2348-8387, Volume 8 Issue 1, 30-38, January 2021.
- [2020] Sanam Narejo, Saima Siraj, Soomro Bushra, Aqsa Noreen, Nadia Memon, "Development of Vehicle Driver Drowsiness detection system using Eye Aspect Ratio", Palarch's Journal Of Archaeology Of Egypt/Egyptology 17(9), 2020. ISSN 1567- 214x. (Scopus).
- [2019] B. N. SOOMRO, N. A. JAFFAR*, S. BHATTI, L. A. THEBO; "A Unique Spectral Spatial Bayesian Framework via Elastic Net Regression for the Classification of Hyperspectral images", Sindh Univ. Res. Jour. (Sci. Ser.), ('Y' category), Vol. 51 (03), 555-564 (2019).
- [2019] Muhammad Moazzam Jawaaid, Bushra Naz, Sajjad Ali Memon, Noor-ur-Zaman Leghari, "Effective Image Segmentation Using Composite Energy Metric in Levelset Based Curve Evolution", Mehran University Research Journal of Engineering and Technology (HEC Recognized "X" Category Journal), Vol. 38, No. 1, January 2019, pp- 175-184.
- [2019] M. MANSOOR, B. NAZ, R. JAFFARI, A. ANSARI, "Brain Controlled Wheelchair with Neuro sky Mind wave"; Sindh Univ. Res. Jour. (Sci. Ser.), ('Y' category), Vol. 51 (04) 709-712 (2019) .), ('Y' category).

- [2018] Bushra Naz, Liang Xiao, Muhammad Moazzam Jawaid, Shahzad Hyder Soomro, Naveed Ahmed Jaffari, "A Bilateral Filter Based Post-Processing Approach for Supervised Spectral- Spatial Hyperspectral Image classification", Mehran University Research Journal of Engineering and Technology (Category: X), Vol: 37 No: 3 (2018): July Issue: <https://doi.org/10.22581/muet1982.1803>
- [2018] Le Sun; Byeungwoo Jeon; Bushra Naz; Yuhui Zheng; Zebin Wu; Liang Xiao, "Fast Superpixel Based Subspace Low Rank Learning Method for Hyperspectral Denoising", IEEE Access (W category: IF: 3.557), Volume: 6, February 2018, pp- 12031 – 12043.
- [2018] Sohni Muneer, Tariq Jamil Khanzada, Samiullah Jatoi, Bushra Naz, "Detection of noisy blood images indicating prodromal diseases", International Journal of Latest Trends in Engineering and Technology, Special Issue ICRMR-2018, Vol (12), Issue (6), pp. 00-006.
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- [2016] Pengfei Liu; Liang Xiao; Jun Zhang; Bushra Naz, Spatial-Hessian-Feature-Guided Variational Model for Pan-Sharpener, IEEE Transactions on Geo science and Remote Sensing (W category IF:5.85), Volume: 54 , Issue: 4 , April 2016, pp-2235 – 2253.
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International / National Conferences

- [C1] K. Batool, **B. Naz**, S. Zai, K. Agha and H. Mahboob, "Bimodal Emotion Recognition: A Deep Multimodal Approach Using Audio and Visual Signals," *2026 1st International Conference on Innovations in Information and Communication Technologies (IICT)*, Jamshoro, Pakistan, 2026, pp. 1-6, doi: 10.1109/IICT68374.2026.11468793.
- [C2] K.Fatima, A.Ahmed, M.Ovais, **N. Bushra**, "An improved IOT based Water quality detection device with Bacterial Prediction, 4th International Conference on Computational Science and Technology, Pakistan, June 2025.
- [C3] **Bushra Naz Soomro**; Liang Xiao; Shahzad Hyder Soomro, "Non-local Spectral-spatial Centralized Sparse Representation for hyperspectral image classification"; 2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Beijing CHINA.
- [C4] S. Ali, **B. Naz**, S. Narejo, S. Ali and J. K. Pabani, "Transformers Unveiled: A Comprehensive Evaluation of Emotion Detection in Text Transcription," *2024 Global Conference on Wireless and Optical Technologies (GCWOT)*, Malaga, Spain, 2024, pp. 1-7, doi:10.1109/gcwot63882.2024.10805688.
- [C5] Kanwal Batool, **Bushra Naz**, Sammar Zai, Kashaf Agha, "Bimodal Emotion Recognition: A Deep multimodal Approach using Audio and Visual Signals", *1st International Conference on Innovations in Information and Communication Technologies (IICT'26)*, January 15-17 2026.
- [C6] Amir Mal, **Bushra Naz**, Shahnawaz Talpur, "Explainable AI Approach for Deep-Learning-Based EMG Signal Classification for Prosthetic Devices", *8th International Multi-Topics*

ICT (IMTIC'25) Conference, December 29-31, 2025.

- [C7] Shahzad Hyder Soomro ; Liang Xiao ; **Bushra Naz** , “Hyperspectral image classification via Elastic Net Regression and bilateral filtering”, 2015 IEEE International Conference on Progress in Informatics and Computing (PIC), 18-20 Dec. 2015, Nanjing China
- [C8] Sabit Rahim ; Sun Tie ; Afsana Begum ; **Bushra Naz**; “ICTs based crime control model: An application based study of Gilgit-Baltistan, Pakistan”; IEEE 2011 10th International Conference on Electronic Measurement & Instruments
- [C9] Sofia Hajano, **Bushra Naz**, Shahnawaz Talpur, “Area and Feature Based Image Registration Using Template Matching and SURF Algorithm”, International Conference on Computational Sciences and Technologies, 17-19 December 2020 (INCCST'20), MUET Jamshoro.
- [C10] Nisha Kumari, Tariq Jameel, **Bushra Naz**, “Design and Deployment of hybrid Indoor positioning System via GPS and WIFI”, International Conference on Computational Sciences and technologies, 2019 (INCCST'19), MUET Jamshoro.
- [C11] Nayab Bashir, Sanam Narejo, **Bushra Naz**, “EEG based Major Depressive Disorder (MDD) detection using Machine Learning”, 5th international conference on Pattern recognition and artificial Intelligence (MedPRAI2021)”, December 17-18, 2021.
- [C12] Songze Tang ; Liang Xiao ; **Bushra Naz** ; Pengfei Liu ; Yufeng Chen; “Joint dictionary learning with ridge regression for pan sharpening”; 2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 26-31 July 2015, Milan, Italy.
- [C13] T J S Khanzada ; A Mukhtiar ; **Naz Bushra** ; M S N Talpur ; A Faisal, “Evaluation and analysis of network coding at network layer”; 2017 International Conference on Progress in Informatics and Computing (PIC), 15-17 Dec. 2017, Nanjing China.